IMO

Verified Knowledge on Chain

A Decentralised Education Protocol for Africa

Token: IMO · Network: Ethereum / 0G Labs · Identity: ZK-SNARKs

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Participation in any token offering carries risk. Readers should conduct their own research and consult qualified advisors before making any financial decisions. The Imo team makes no representations or warranties regarding the accuracy of projections or forward-looking statements contained in this document.

Abstract

Imo is a decentralised education protocol built on Ethereum and 0G Labs. It makes academic credentials verifiable, portable, and economically meaningful for students, educators, and institutions across Africa. The protocol combines on-chain credential issuance through soulbound NFTs, Zero-Knowledge Proof identity verification, decentralised file storage, and a culturally grounded AI layer with native support for Yoruba, Hausa, and Igbo.

The platform is governed by the IMO token — an ERC-20 utility token that rewards verified learning, enables platform access, and powers community governance. Imo is designed to serve the 40+ million students currently enrolled in Nigerian educational institutions, and to extend reach to the millions more who learn informally through trade apprenticeships and community programmes — but currently hold no verifiable record of what they know.

> **Core Thesis** The African education system does not lack learners. It lacks infrastructure to make learning legible. Imo is that infrastructure — a trust layer for knowledge, built on blockchain, powered by AI, and owned by the community it serves.

1. The Problem

1.1 The Invisible Graduate

Every year, millions of students across Nigeria complete years of education, pass examinations, develop real skills, and leave with a paper certificate the world cannot verify. A graduate from the University of Ilorin cannot instantly prove their degree to an employer in London. A student who completed vocational training in Kano carries nothing portable or trusted. A community health worker who trained through an NGO programme holds nothing but a piece of paper anyone could forge.

This is the invisible graduate problem. The knowledge is real. The effort is real. But the record is fragile, forgeable, and worthless the moment it leaves the issuing institution's filing cabinet.

1.2 Scale of the Crisis

Metric	Figure	Source
Students enrolled in Nigerian tertiary education	Over 2 million	NUC, 2023
Secondary school students in Nigeria	Over 14 million	UBEC, 2023
Certificate forgery — documented cases	Widespread; estimated 30–40% of employers report suspicious credentials	Employer surveys, NUC
Students without reliable internet access	Over 60% in rural areas	NCC, 2023
Languages spoken in Nigeria	Over 500	SIL Ethnologue
EdTech platforms with Yoruba/Hausa/Igbo support	Near zero	Market research, 2024

1.3 Why Existing Solutions Fail

Several attempts have been made to digitise educational records in Nigeria. They share a common failure: they replace one centralised system with another.

A government database can be hacked, corrupted, or taken offline by a change in administration. A company-hosted credential platform can shut down, taking all student records with it. Platforms like Blockcerts and Awoke offer partial blockchain solutions but were not designed for the African context. They assume reliable internet, smartphone ownership, English literacy, and institutional infrastructure that does not exist across much of Nigeria.

The fundamental problem is not technological. It is architectural. Any solution that depends on a single institution, company, or government to remain trustworthy is not a solution to the trust problem. It is a deferral of it.

1.4 The Five Root Failures

- No cryptographic proof of credential authenticity. Paper certificates carry no unforgeable signature.
- No decentralised storage. Student records depend on institutional servers that can fail, be seized, or be deleted.
- No language accessibility. Educational platforms are English-only in a country where over 60% of the population primarily speaks an indigenous language.
- No economic incentive for sustained learning. Students in low-income communities have no tangible reward for consistent educational engagement.
- No privacy-preserving identity. Students cannot selectively prove credentials without exposing their full personal data.

2. The Solution — Imo Protocol

2.1 What Imo Is

Imo is a decentralised education protocol. Not an app. Not a product students depend on a company to maintain. It is an open protocol — a set of smart contracts, storage proofs, and AI services — that any institution, educator, or student can build on, interact with, and trust without needing to trust any single counterparty.

At its core, Imo makes four promises:

- Your credentials are yours. Minted as soulbound NFTs on Ethereum, they cannot be transferred, forged, or revoked by anyone other than the issuing institution under strict governance rules.
- Your identity is private. Zero-Knowledge Proofs allow you to prove who you are and what you know without revealing personal data to any third party.
- Your records are permanent. Files stored on 0G Labs carry on-chain existence proofs. They cannot be lost by institutional failure, government action, or company closure.
- Your learning has value. Every verified learning milestone earns IMO tokens — a real economic reward for real intellectual work.

2.2 The Protocol Stack

Layer	Component	What It Does
Trust Layer	**Ethereum Smart Contracts**	Immutable credentials · token rewards · governance
Identity Layer	**ZK-SNARKs via Circom**	Prove identity without revealing personal data
Storage Layer	**0G Labs**	Decentralised, verifiable document storage
Intelligence Layer	**Multilingual LLM**	Yoruba · Hausa · Igbo · English AI services
Access Layer	**React PWA + USSD**	Web · Mobile · Feature-phone · Offline
Economy Layer	**IMO ERC-20 Token**	Rewards · Governance · Platform payments

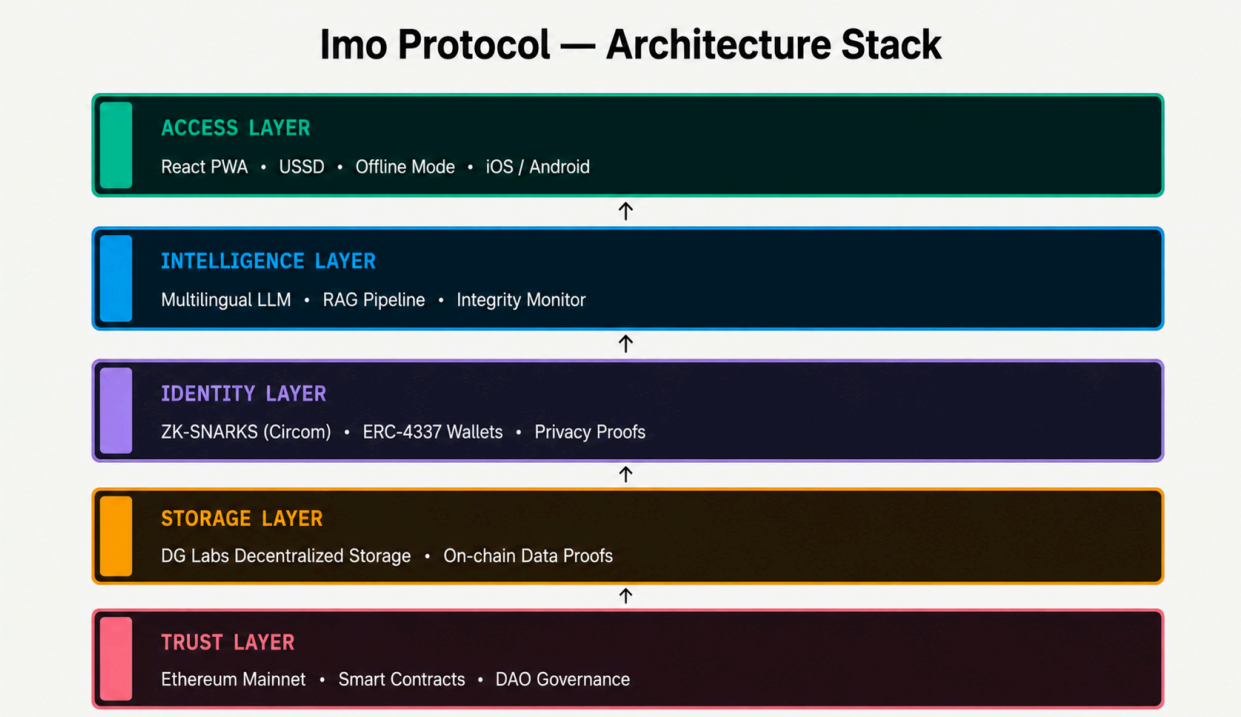


Figure 1 — Imo Protocol Architecture Stack

2.3 Plain Language: How ZK Identity Works

Think of it like showing a bouncer your ID to prove you are over 18. They confirm the fact without reading your home address, your date of birth, or any other detail you did not choose to

share. The bouncer learns one thing: you qualify. Imo's ZK-SNARK layer works the same way — an employer learns that your credential is real and that you are who you say you are. They learn nothing else.

2.4 Plain Language: How 0G Labs Storage Works

Every file stored through Imo receives a cryptographic fingerprint — a unique hash recorded on-chain. If anyone alters the file in any way, the fingerprint changes, and the mismatch is immediately visible to anyone who checks. It is a tamper-evident seal on every document, backed by mathematics rather than institutional promises.

2.5 The Cultural Layer

Most blockchain and AI education platforms are built in the Global North and adapted — poorly — for African markets. Imo is built in Africa, for Africa, from the ground up. The most visible expression of this is the language layer: Imo natively supports Yoruba, Hausa, and Igbo as first-class languages.

Not as translation plugins. Not as optional add-ons. As core architectural decisions baked into the AI model, the interface design, and the user onboarding flow. A student in Kano receives the full Imo experience in Hausa. A student in Ibadan navigates the platform in Yoruba. A student in Enugu interacts with the AI chatbot in Igbo. This is not a feature. It is a statement of who this platform was built for.

3. Protocol Architecture

3.1 Smart Contract Layer

ImoCredential.sol — Soulbound NFT Credentials

The credential contract implements the ERC-721 standard with a permanent soulbound modification: the transfer function is disabled at the contract level. Certificates minted through this contract are bound to the recipient's wallet address for life. The contract stores the credential metadata hash, the 0G Labs content URI for the full certificate document, the issuing institution's verified on-chain address, and a revocation flag controlled by a multi-sig governance process.

Verification is gasless. Any employer or institution can call the verify() function with a credential ID and receive a cryptographic confirmation of authenticity without paying gas. This is critical for adoption in markets where gas cost is a barrier.

ImoToken.sol — ERC-20 Utility Token

The IMO token is the economic engine of the protocol. It is minted according to a fixed reward schedule triggered by verified on-chain learning events. Reward events include course completion (verified by educator signature), assessment pass (verified by malpractice-clean assessment hash), credential issuance, and consistent learning streaks at 30, 60, and 90-day intervals. Tokens are burned when spent on platform services, creating deflationary pressure as adoption grows.

ImoAccess.sol — Role-Based Access Control

On-chain RBAC defines four roles: STUDENT, EDUCATOR, ADMIN, and PARENT. Each role carries precisely scoped permissions. STUDENT can read their own records and earn tokens. EDUCATOR can issue credentials to enrolled students. ADMIN can assign roles and manage institutional parameters. PARENT holds read-only access to their linked student's activity data. Role assignments require multi-signature approval and are publicly auditable on-chain.

ImoGovernance.sol — DAO

Protocol parameters — reward rates, fee structures, new feature approvals, institutional onboarding requirements — are controlled by a DAO governed by IMO token holders. Proposals require a minimum token threshold to submit. Voting is time-weighted: tokens must be held for at least 30 days to carry full voting weight, preventing governance attacks by short-term token accumulation.

3.2 Zero-Knowledge Identity Layer

Standard blockchain identity requires users to link a real-world identity to a wallet address — a privacy violation that exposes students to tracking and discrimination. Imo uses ZK-SNARKs to allow students to prove identity without revealing it.

The ZK circuit, built with Circom, takes three private inputs: a hash of the student's government-issued ID number, a hash of their date of birth, and a nonce. It produces a single public output: a proof stored on-chain alongside the student's wallet address. When an employer queries a credential, the verification confirms that this wallet is associated with a verified unique human identity — nothing more. The employer learns the credential is real. They learn nothing else.

3.3 Storage Layer — 0G Labs

0G Labs provides programmable decentralised storage with on-chain data availability proofs. Every file stored through Imo — certificates, course materials, student portfolios, assessment recordings — is assigned a content hash recorded on-chain. Tampering is mathematically detectable.

0G Labs was chosen over IPFS for three specific reasons: guaranteed data availability without pinning services, programmable storage with smart contract integration, and lower storage costs for the high-volume document use case that education requires.

3.4 AI Layer

Multilingual Intelligence

The AI layer is built on a fine-tuned multilingual base model with additional training on Yoruba, Hausa, and Igbo language corpora sourced from the Masakhane research project, university linguistics departments, and community-contributed datasets. The model supports full interface translation, AI chatbot conversations, document summarisation, and assessment generation across all three languages.

Document Intelligence — RAG Pipeline

The Retrieval-Augmented Generation pipeline processes uploaded educational documents: PDFs, lecture notes, textbooks, and research papers. The pipeline extracts text, generates semantic vector embeddings, stores them in a vector database, and on query retrieves relevant chunks to pass to the LLM for synthesis. The output is a structured concept diagram, a summary, or a direct answer, depending on the student's query.

Academic Integrity Monitor

The malpractice detection system runs as a lightweight client-side model during assessments. It monitors clipboard events, tab visibility, screen sharing activity, and keystroke patterns for anomalies. All events are cryptographically signed and logged with timestamps. At assessment completion, the integrity log is hashed and submitted on-chain alongside the result — an immutable audit trail of assessment conditions.

4. Token Economics

4.1 IMO Token Overview

Parameter	Value
Token name	IMO
Token standard	ERC-20
Total supply	1,000,000,000 IMO (1 billion, fixed)
Network	Ethereum mainnet (Polygon L2 in Phase 2)
Decimals	18

Parameter	Value
Utility	Platform access, learning rewards, governance voting
Deflationary mechanism	50% burn on all platform service spend

4.2 Token Distribution

	Allocation	%	Amount (IMO)	Vesting Schedule
	Community Rewards Pool	**40%**	400,000,000 IMO	*Released over 10 years per learning milestones*
	Ecosystem Development	**20%**	200,000,000 IMO	*4-year linear vest, 1-year cliff*
	Team & Founders	**15%**	150,000,000 IMO	*4-year linear vest, 1-year cliff*
	Institutional Partners	**10%**	100,000,000 IMO	*2-year linear vest, 6-month cliff*
	Treasury Reserve	**10%**	100,000,000 IMO	*DAO-controlled, milestone-gated*
	Public Sale	**5%**	50,000,000 IMO	*No vesting — liquid at listing*

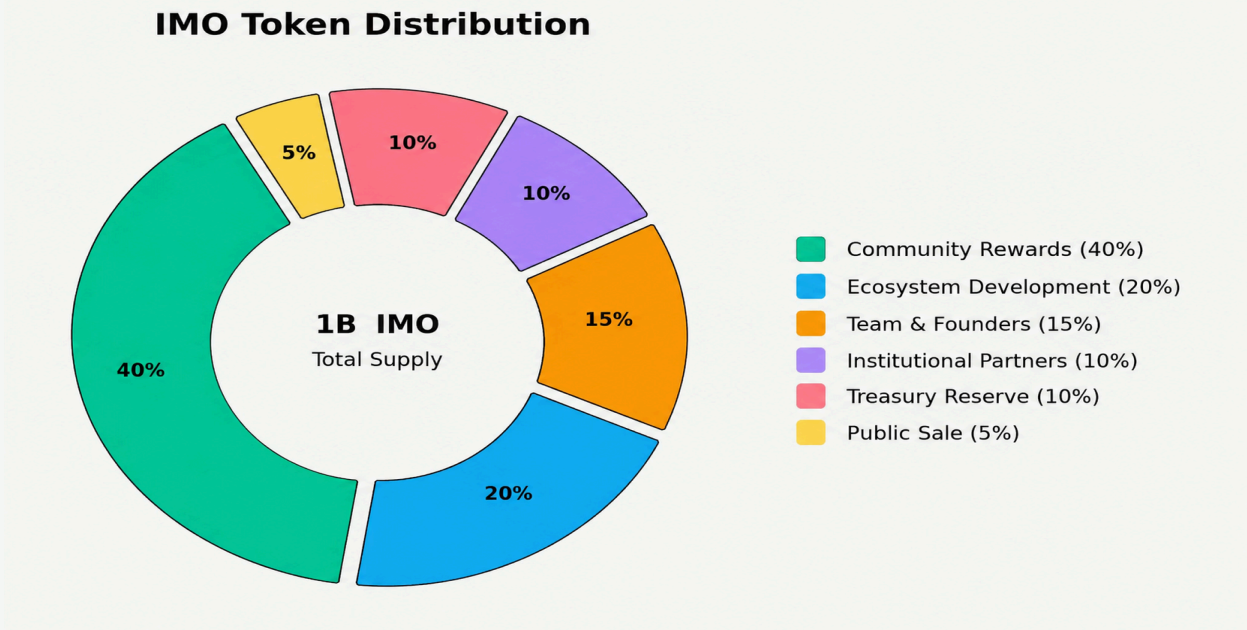


Figure 2 — IMO Token Distribution (1B Fixed Supply)

4.3 Emission Schedule

Period	Annual Emission	Scale	Triggering Events
Year 1–2	**80,000,000 IMO / year**		*Course completions · credentials issued · streak bonuses*
Year 3–4	**40,000,000 IMO / year**		*Same events, halved emission rate*
Year 5–6	**20,000,000 IMO / year**		*Same events, halved again*
Year 7–10	**10,000,000 IMO / year**		*Tail emissions for sustained ecosystem activity*

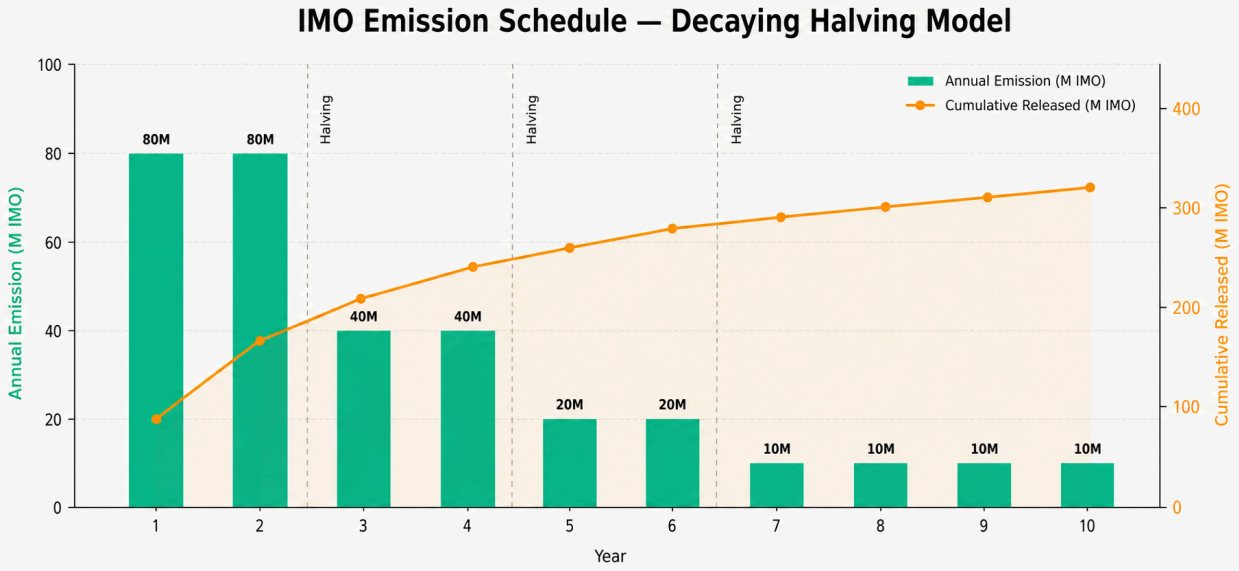


Figure 3 — IMO Emission Schedule: Decaying Halving Model

4.4 Why Hold IMO for 12+ Months?

This is the question every serious investor will ask. The answer must be mechanical, not narrative.

Short-term token holders face two structural disincentives. First, all platform service payments burn 50% of the tokens spent — meaning token scarcity increases as platform usage grows.

Second, governance voting requires a 30-day minimum hold for full weight, which means tokens traded frequently carry reduced influence.

The long-term hold thesis is built on three mechanics:

- Staking yield tied to protocol revenue. Staked IMO tokens earn a proportional share of credential verification fees collected from employers and institutions. As more institutions join the network and issue credentials, the verification fee pool grows. Stakers benefit directly from protocol adoption — not from price speculation.
- Milestone-gated treasury releases. Students who accumulate IMO rewards can lock tokens in the treasury staking pool for 12-month periods. Locked tokens receive a 1.25x multiplier on governance voting weight and a share of marketplace revenue. This creates a direct economic reason for students — the protocol's primary users — to hold rather than sell.
- Network effect compounding. Every institution that joins Imo makes credentials from all other institutions more valuable, because employers learn one verification interface. Token holders who hold through the network's growth phase capture the value of that compounding directly.

4.5 Token Utility

Earn

- Complete a verified course — 50 IMO
- Pass a malpractice-clean assessment — 20 IMO
- Receive a credential issuance — 100 IMO
- 30-day learning streak — 75 IMO bonus
- Refer an institution to the protocol — 500 IMO

Spend

- Purchase premium course materials — IMO payment, 50% burned
- Access AI chatbot premium features — IMO subscription, 50% burned
- Pay for book purchases via platform marketplace — IMO or ETH
- Submit governance proposals — 1,000 IMO minimum, returned if proposal passes

Stake

- 12-month lock — earn share of credential verification fee revenue
- 1.25x governance weight multiplier for locked tokens
- Early unstake penalty: 10% of staked amount, redistributed to remaining stakers

4.6 Treasury Milestone Gates

The 100,000,000 IMO treasury reserve is not a blank cheque. Releases are tied to verifiable on-chain and off-chain milestones. The following gate structure applies:

Gate	Condition	Release Amount
Gate 1	500 students onboarded to Landmark pilot with verified credentials on-chain	10,000,000 IMO
Gate 2	First Ministry of Education MOU signed and publicly announced	15,000,000 IMO
Gate 3	10,000 verified credentials issued across 3+ institutions	20,000,000 IMO
Gate 4	IMO token mainnet deployment with smart contract audit passed	20,000,000 IMO
Gate 5+	Remaining 35M governed by DAO vote after Gate 4 is reached	35,000,000 IMO

> ⚠ **Note** Treasury milestone gates are enforced at the smart contract level. No single team member or multi-sig can release treasury funds outside of verified gate conditions. All gate verifications are published on-chain with supporting documentation.

4.7 Deflationary Mechanics

Every IMO token spent on platform services is split: 50% goes to the protocol treasury and 50% is permanently burned. As platform adoption grows, the burn rate increases. Combined with the decaying emission schedule, this creates sustained deflationary pressure on token supply over time. The model is designed to ensure long-term token holders — students who accumulate rewards — benefit from growing platform adoption.

5. Product

5.1 Core Platform Features

Feature	Description	Technology
Soulbound credential NFTs	Immutable on-chain certificates tied to student wallet	ImoCredential.sol / ERC-721
ZK-proof identity	Prove credentials without revealing personal data	Circom / SnarkJS
Decentralised storage	All files stored on 0G Labs with on-chain proofs	0G Labs protocol
Role-based access	On-chain RBAC for Students, Educators, Admins, Parents	ImoAccess.sol
Indigenous language support	Full platform in Yoruba, Hausa, and Igbo	Fine-tuned LLM
Document intelligence	Upload any document, get AI-generated concept diagram	RAG + LLM pipeline
AI chatbot with voice	Context-aware chatbot, voice overview of any document	LLM + voice synthesis
Malpractice detection	Real-time monitoring during assessments	Computer vision AI
IMO token rewards	Earn tokens for every verified learning milestone	ImoToken.sol / ERC-20
Parent engagement	Weekly SMS reports on student activity	USSD gateway + ImoAccess.sol
Book marketplace	Buy and sell course materials via ETH or IMO	Smart contract payments

5.2 User Journeys

The Student — Amina, Kano

Amina is 19, studying business administration at a polytechnic in Kano. Her first language is Hausa. She registers on Imo, creates a gasless wallet via ERC-4337 account abstraction, and completes identity verification using a ZK proof tied to her student ID. She uploads her course textbook as a PDF and immediately receives a concept diagram of the first chapter. She completes her semester and her institution issues her results on-chain. She earns 200 IMO tokens.

When she graduates and applies for a job in Lagos, her employer verifies her credentials in ten seconds — without calling the university, without waiting for a transcript, and without Amina sharing anything beyond a credential ID.

The Educator — Dr. Adeyemi, Ibadan

Dr. Adeyemi teaches computer science at a federal university in Ibadan. He registers as an educator on Imo, passing the subject-matter assessment that verifies his teaching competence on-chain. He creates a course, uploads materials to 0G Labs, and sets an assessment. When 30 students complete the course, he reviews their integrity logs and issues credentials to students who passed with clean assessment records. Each issuance costs a small gas fee — subsidised by the protocol treasury for the first 10,000 credentials issued under the institutional onboarding programme.

The Parent — Ngozi, Enugu

Ngozi's daughter Adaeze is in secondary school in Enugu. Ngozi does not own a smartphone. She receives weekly SMS notifications via USSD gateway summarising Adaeze's learning activity, assessment scores, and earned tokens. When Adaeze's activity rating drops below 60%, Ngozi receives an immediate alert. She calls the school. The problem is identified early. This is the feedback loop Nigerian education has never had.

The Institution — Landmark University, Omu-Aran

Landmark University integrates the Imo protocol via the open API. Their existing student management system maps to Imo's on-chain identity layer. When a student graduates, the registry issues a credential NFT in addition to the paper certificate. The university pays a flat annual protocol fee in IMO tokens. Within two years, every Landmark graduate holds a verifiable, portable, globally accessible academic record. Forgery becomes impossible.

6. Go-to-Market Strategy

6.1 Beachhead — One Institution, One City

Imo will not launch everywhere at once. The beachhead strategy targets one institution with a complete, proven implementation before expanding. The target is Landmark University, Omu-Aran, Kwara State — a private university with a forward-thinking administration, a strong STEM programme, and an existing relationship with the Imo founding team. A full pilot covering student enrollment, credential issuance, and parent engagement will run across one academic semester.

6.2 Expansion Phases

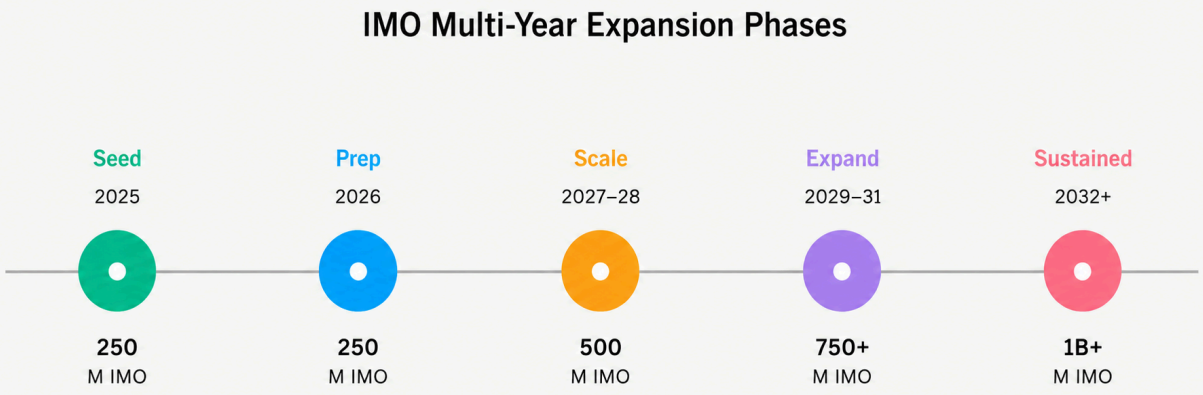


Figure 4 — Go-to-Market Expansion Timeline

	Phase	Timeline	Target	Goal
	Phase 0 — Build	Now – Month 6	Core team · STEM Festival	*Prototype · competition entry · initial funding*
	Phase 1 — Pilot	Month 7–12	Landmark University	*Full semester pilot · 500+ students on-chain*
	Phase 2 — State	Month 13–24	Kwara State institutions	*10+ institutions · 50,000+ students*
	Phase 3 — National	Month 25–36	5 Nigerian states	*Ministry partnerships · 500,000+ students*
	Phase 4 — Continental	Month 37–48	Ghana · Kenya · Rwanda	*Pan-African expansion via protocol licensing*

6.3 Distribution Channels

- Direct institutional sales — university and polytechnic administrations
- Ministry of Education partnerships — federal and state-level policy integration
- NGO and donor-funded deployments — USAID, Gates Foundation, Tony Elumelu Foundation
- Student-led adoption — IMO token rewards create organic peer-recruitment incentive
- Developer ecosystem — open API and SDK for third-party EdTech integrations

6.4 Revenue Model

Revenue Stream	Model	Pricing
Institutional subscription	Annual flat fee per institution	Tiered by student enrollment size
Credential issuance	Per-credential fee (gas + protocol fee)	Subsidised for first 10,000 per institution
AI features	Premium subscription in IMO tokens	Monthly or annual, burnable
Marketplace commission	5% of all transactions	IMO or ETH, 50% burned
Verification API	Per-query pricing for employer verification	Volume discounts available
Protocol licensing	White-label deployments for national ministries	Custom enterprise pricing

7. Team

7.1 Founding Team

Imo was conceived at Landmark University in Omu-Aran, Nigeria, by a team of builders, researchers, and educators who have experienced the failures of the African education system firsthand — as students, as graduates who struggled to have credentials recognised internationally, and as builders who have spent years working on blockchain and AI infrastructure. Every member of the team has a personal stake in seeing this protocol succeed.

Role	Background	Verifiable Credentials
Co-Founder & CEO	[Name] — [University, Year]Previous: [Company / Project]	[LinkedIn URL] · [GitHub / Portfolio]
Co-Founder & CTO	[Name] — [University, Year]Solidity / ZK-SNARK expertise	[LinkedIn URL] · [GitHub / Portfolio]
Head of AI / NLP	[Name] — [University, Year]African language NLP background	[LinkedIn URL] · [Research Publications]
Head of Partnerships	[Name]Nigerian institutional relationships	[LinkedIn URL] · [Past Institutional Deals]
Legal & Compliance	[Name / Firm]Nigeria SEC, NDPA expertise	[Bar number / Firm URL]

7.2 Advisors

The protocol benefits from advisory relationships in the following domains. Specific advisor names and affiliations to be inserted prior to investor distribution:

- Blockchain infrastructure — smart contract security, EVM architecture
- African EdTech — institutional adoption strategy, ministry relationships
- Token economics — mechanism design, vesting structure validation
- African linguistics — Yoruba, Hausa, and Igbo language model supervision

7.3 Protocol Governance Transition

Imo launches as a founder-led project with a clear path to full DAO governance across three phases:

Phase	Governance Model	Timeline
Foundation	Core team holds admin keys with a 2-of-3 multi-sig. All decisions documented publicly.	Year 1
Community	DAO governance activated for non-critical parameters. Token holders vote on reward rates, new institutional partners, and fee structures.	Year 2
Full Decentralisation	All protocol parameters governed by DAO. Core team retains technical advisory role only. Protocol council elected annually.	Year 3+

8. Competitive Landscape

8.1 Comparison

The table below includes Imo's own current limitations. Presenting an honest competitive picture is more useful to investors than a clean sweep.

Capability	Imo	Awoke	Blockcerts	Learning Economy	Coursera
On-chain credentials	Yes	Partial	Yes	Yes	No
ZK-proof identity	Yes	No	No	No	No

Capability	Imo	Awoke	Blockcerts	Learning Economy	Coursera
African language support	Yes	No	No	No	No
Decentralised storage	Yes (0G)	No	Partial	No	No
Token reward economy	Yes	No	No	Partial	No
AI document intelligence	Yes	No	No	No	No
Malpractice detection	Yes	No	No	No	No
Offline / USSD access	Yes	No	No	No	No
Global brand recognition	No (early stage)	No	Partial	No	Yes
Enterprise partnerships	No (early stage)	No	Yes	Partial	Yes
Built for Africa	Yes	Partial	No	No	No

8.2 Where Incumbents Have the Advantage

Honesty about current weaknesses is more persuasive than a clean sweep. Two things incumbents do better than Imo right now:

- Coursera and established global platforms have years of brand trust with international employers. An Imo credential is unknown to employers outside Nigeria in Year 1. This changes as network adoption grows — but it is a real friction point in the early phases.
- Blockcerts has existing enterprise partnerships with universities in Europe and North America. Their institutional sales pipeline is more mature. Imo's advantage is being architecturally designed for the African context from the start; Blockcerts retrofitting their stack for Nigeria would require rebuilding core assumptions.

8.3 Sustainable Moat — Structural Arguments

The question every investor asks: why can't a well-resourced incumbent just copy this?

Three structural mechanisms make replication genuinely difficult, not just temporarily slow:

First, community token alignment with low-income users is an architectural decision, not a feature. Coursera's business model is built on premium subscriptions from users in high-income markets. Aligning their token economics with Nigerian students on 50 IMO per course completion would require a fundamental business model rewrite that conflicts with their existing investor obligations.

Second, the Yoruba, Hausa, and Igbo training datasets represent years of academic linguistic work. A new entrant cannot replicate this with a translation API. The Masakhane datasets required community partnerships, academic review, and cultural calibration that money alone cannot accelerate.

Third, institutional trust in Africa is relationship-driven, not product-driven. The reason Landmark University is the beachhead is that the founding team has an existing relationship there. A well-funded competitor cannot buy that relationship; they would need to build it from scratch over 12–18 months, by which time Imo has credentials issued, network effects established, and verified students the competitor's product has never reached.

9. Risks and Mitigations

Naming risks before an investor does is not weakness. It is the fastest way to demonstrate that the team understands the space well enough to have thought through the hard problems.

Risk	Why It Matters	Specific Mitigation
Smart contract vulnerability	A single exploit could drain the community rewards pool or corrupt credential records — destroying user trust overnight.	Formal audit before every mainnet deployment. Bug bounty programme. Multi-sig admin on all critical functions.
Regulatory action on IMO token (Nigeria SEC)	Nigeria's SEC has been unpredictable on crypto assets. An adverse ruling could restrict IMO trading or force a restructure. This is not a low-probability risk.	Obtain formal legal opinion from qualified Nigerian counsel before token launch. Structure IMO as a pure utility token with no profit-sharing mechanism. Engage SEC proactively with whitepaper and legal memo prior to public sale.
Low institutional adoption	The protocol is worthless without institutions issuing credentials. If Landmark University delays or exits the pilot, the entire go-to-market collapses.	Secure a signed MOU with Landmark before the pilot start date. Build two backup institutional relationships in parallel. Design the pilot so it delivers value to the institution regardless of full protocol deployment.
Language model accuracy	Mistranslation of assessment questions or credential content in indigenous languages would undermine the platform's core cultural value proposition.	Human review layer for all assessment content. Community feedback loop with native speakers. Quarterly model audits by linguistics advisors.
Internet connectivity barriers	Over 60% of target users are in low-connectivity environments. A	Offline-first PWA with local data sync. USSD fallback for all critical

Risk	Why It Matters	Specific Mitigation
	product that requires broadband is not a product for this market.	functions. SMS notification system for parents.
Gas cost barriers	Ethereum mainnet gas costs are prohibitive for low-income users. A student who earns 50 IMO cannot spend 60 IMO in gas to claim it.	Gasless wallets via ERC-4337 account abstraction. Gas subsidisation for credential issuance in Phase 1. Polygon L2 migration in Phase 2 reduces gas costs by approximately 95%.
Team key person risk	If the founding CTO exits before mainnet, the ZK-SNARK implementation is at risk.	Full technical documentation maintained in real time. Knowledge distributed across at least two senior engineers. Succession protocol documented and agreed by co-founders.

10. Fundraising & Use of Funds

10.1 Seed Round

Parameter	Value
Round type	Pre-Seed / Seed
Target raise	$250,000 – $500,000
Token instrument	SAFE with token warrant (pro-rata rights on public sale allocation)
Target close	July 2025
Use of proceeds	18 months of runway through testnet launch and Landmark pilot

10.2 FDV Scenario Analysis

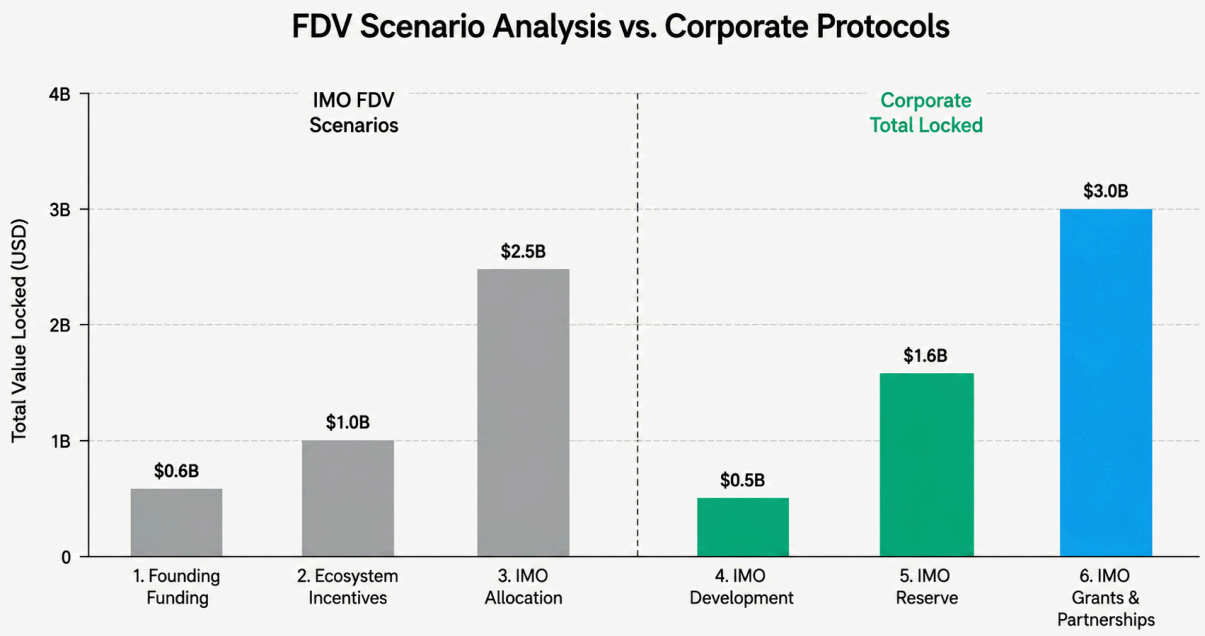


Figure 5 — IMO FDV Scenarios vs. Comparable Protocols at Seed Stage

The conservative scenario ($10M FDV) assumes a $0.01 launch price with limited initial liquidity. The base case ($30M FDV) aligns with comparable African blockchain protocols at seed stage. The optimistic case ($80M FDV) reflects comparable Web3 education protocols that have demonstrated early institutional traction before public token launch. These are not price targets. They are scenario benchmarks to contextualise valuation expectations.

Scenario	FDV	Relative Scale	Context
Conservative	**$10M FDV**		**$0.01 launch price** *Limited initial liquidity · early-stage traction only*
Base Case	**$30M FDV**		**$0.03 launch price** *Comparable African blockchain protocols at seed stage*
Optimistic	**$80M FDV**		**$0.08 launch price** *Web3 education protocols with demonstrated institutional traction*

10.3 Use of Funds

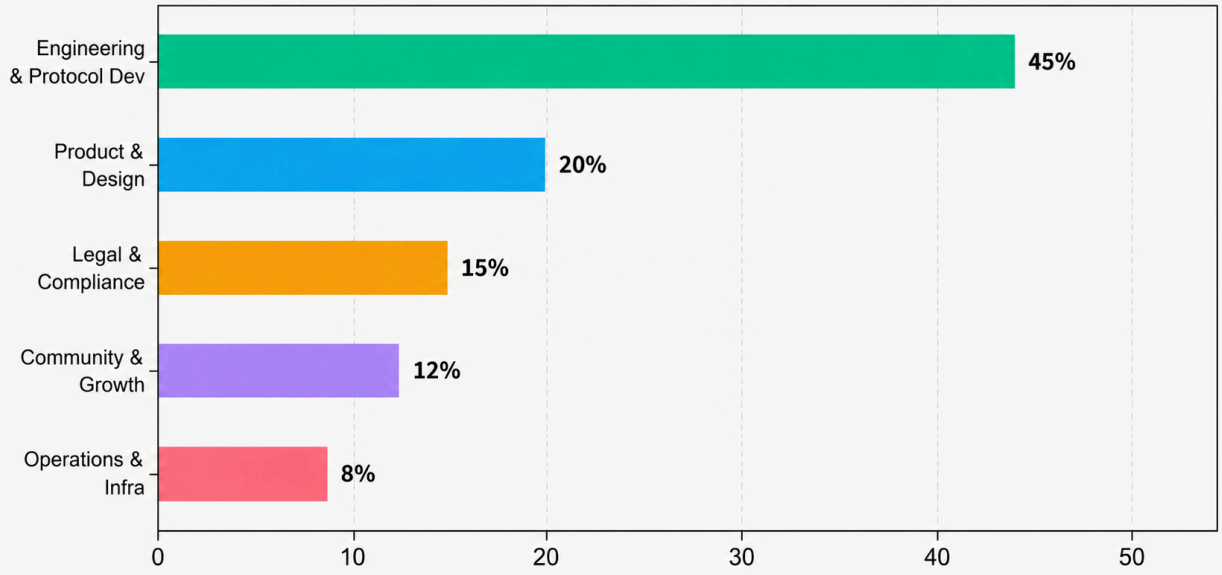


Figure 6 — Seed Round Allocation ($250K – $500K)

	Category	%	Scale	What It Funds
	Engineering & Protocol Dev	**45 %**		**$112K – $225K** *Two senior Solidity engineers · ZK-SNARK circuits · smart contract audits*
	Product & Design	**20 %**		**$50K – $100K** *React PWA · USSD integration · AI model fine-tuning*
	Legal & Compliance	**15 %**		**$37K – $75K** *Nigeria SEC legal opinion · token structure review · NDPA compliance*
	Community & Growth	**12 %**		**$30K – $60K** *Landmark University pilot · student onboarding incentives · developer SDK*
	Operations & Infrastructure	**8%**		**$20K – $40K** *oG Labs storage · server infrastructure · team operations*

11. Roadmap

Milestone	Target	Deliverable
STEM Festival submission	May 2025	Competition prototype + written report + video demo
Competition result	June 2025	Award and public launch announcement
Seed funding close	July 2025	$250,000 – $500,000 pre-seed round
Testnet launch	August 2025	All four smart contracts on Sepolia, public testing open
Landmark pilot start	September 2025	500 students onboarded, first credentials issued on-chain
Mobile app launch	October 2025	iOS and Android PWA with offline mode
IMO token generation	December 2025	Token deployed, community rewards pool activated
Mainnet launch	January 2026	Production deployment on Ethereum mainnet
State ministry MOU	March 2026	First state-level Ministry of Education partnership
Polygon L2 migration	June 2026	Gas cost reduction, 10x transaction throughput
Pan-African expansion	2027	Ghana, Kenya, Rwanda protocol deployments

12. Technical Specifications

12.1 Smart Contract Specifications

Contract	Standard	Key Functions	Audit Status
ImoCredential.sol	ERC-721 (soulbound)	mint(), verify(), revoke(), tokenURI()	Pre-launch audit required
ImoToken.sol	ERC-20	mint(), burn(), rewardLearner(), stake(), unstake()	Pre-launch audit required
ImoAccess.sol	Custom RBAC	grantRole(), revokeRole(), hasRole(), checkPermission()	Pre-launch audit required

Contract	Standard	Key Functions	Audit Status
ImoGovernance.sol	Custom DAO	propose(), vote(), execute(), getProposal()	Pre-launch audit required

12.2 Non-Functional Requirements

Requirement	Target
Page load (3G mobile)	Under 3 seconds
Smart contract confirmation (Polygon L2)	Under 2 seconds
AI inference latency	Under 5 seconds for document queries
Platform uptime SLA	99.5%
Concurrent users at launch	100,000+
Data encryption	AES-256 at rest, TLS 1.3 in transit
Backup frequency	Daily automated, geographically distributed
Recovery time objective	Under 4 hours
NDPA compliance	Full — assessed quarterly
Smart contract audit	Before every mainnet deployment

13. Ethical Commitments & Compliance

- Student data sovereignty — students own their data. The protocol will never sell, license, or monetise student personal data under any circumstances.
- Open protocol — all smart contracts are open source and publicly auditable from day one.
- Language equity — Yoruba, Hausa, and Igbo are maintained as permanent first-class languages. Budget and engineering resources are committed to these languages at parity with English.
- No advertising — the platform will never display advertising or allow commercial targeting based on student activity data.
- NDPA compliance — full compliance with Nigeria's Data Protection Act 2023 is maintained and assessed quarterly by qualified legal counsel.
- Community treasury transparency — all treasury transactions are published on-chain and in monthly public reports, regardless of governance phase.

14. Conclusion

Africa does not have an education problem. Africa has an infrastructure problem. The knowledge is there. The students are there. The teachers are there. What is missing is the layer of trust that makes learning legible, portable, and valuable beyond the walls of any single institution.

Imo is that layer. It is built on the most robust trust infrastructure available — Ethereum smart contracts, Zero-Knowledge cryptography, and decentralised storage — and delivered through a culturally grounded experience that speaks the languages, respects the context, and serves the people who have been excluded from every previous wave of educational technology.

The IMO token creates economic alignment between the protocol's success and the success of every student who uses it. When a student earns tokens for learning, they become stakeholders in the protocol that rewarded them. When an institution issues credentials on-chain, they join a network that grows more valuable with every new institution that follows.

> **Core Thesis** Every student who learns something real deserves a record that is as real and as permanent as their knowledge. Imo makes that possible — for every student, in every language, on every device, connected or not, in Lagos or in the village. Knowledge verified. Value created. Ownership returned.

Join us.

Appendix A — Glossary

Term	Definition
Soulbound NFT	A non-fungible token whose transfer function is permanently disabled. It cannot be sold, traded, or moved to another wallet.
ZK-SNARK	Zero-Knowledge Succinct Non-Interactive Argument of Knowledge. A cryptographic proof that demonstrates knowledge of information without revealing the information itself.
ERC-721	Ethereum token standard for non-fungible tokens (NFTs).
ERC-20	Ethereum token standard for fungible tokens such as IMO.
ERC-4337	Ethereum standard for account abstraction — enables wallets without seed phrases, reducing onboarding friction for non-technical users.
0G Labs	A decentralised storage protocol with on-chain data availability proofs.
RAG	Retrieval-Augmented Generation. An AI technique that combines document retrieval with language model generation.
RBAC	Role-Based Access Control. A security model that restricts system access based on user roles.
NDPA	Nigeria Data Protection Act 2023. Nigeria's primary data privacy legislation.
DAO	Decentralised Autonomous Organisation. A governance structure where decisions are made by token holders via on-chain voting.
Circom	A programming language for writing zero-knowledge proof circuits.
SnarkJS	A JavaScript library for generating and verifying ZK-SNARK proofs.
FDV	Fully Diluted Valuation. The market capitalisation of a token assuming all tokens are in circulation.
SAFE	Simple Agreement for Future Equity. A standard early-stage investment instrument.

Appendix B — References

Source	URL	Accessed
Ethereum Foundation	ethereum.org	May 2025
0G Labs Documentation	0g.ai/docs	May 2025
Masakhane African NLP Project	masakhane.io	May 2025
Circom Documentation	docs.circom.io	May 2025
Nigeria Data Protection Act 2023	ndpc.gov.ng	May 2025
National Universities Commission	nuc.edu.ng	May 2025
Universal Basic Education Commission	ubec.gov.ng	May 2025
SIL Ethnologue: Languages of Nigeria	ethnologue.com	May 2025
Blockcerts Open Standard	blockcerts.org	May 2025
EIP-4337: Account Abstraction	eips.ethereum.org/EIPS/eip-4337	May 2025
NCC Digital Report 2023	ncc.gov.ng	May 2025